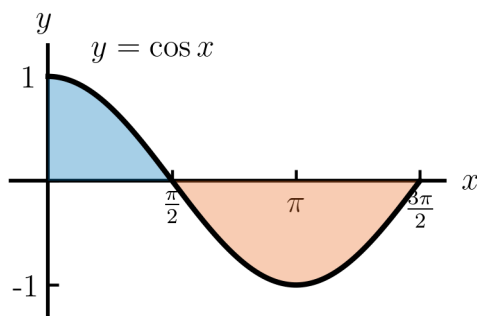


The Definite Integral

In this lesson, we will explore net area bounded by a curve, define the definite integral, and practice matching definite integrals to limits of Riemann sums.

So far, we have approximated the area under a non-negative function f on $[a, b]$. If f takes positive and negative values, then Riemann sums approximate the **net area** bounded by the graph of f and the x -axis.

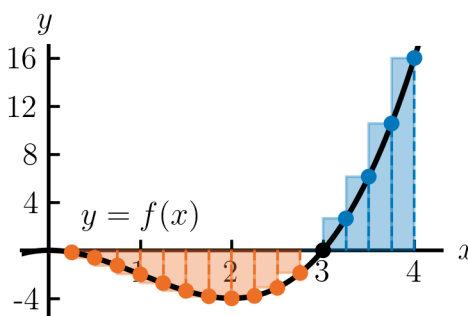
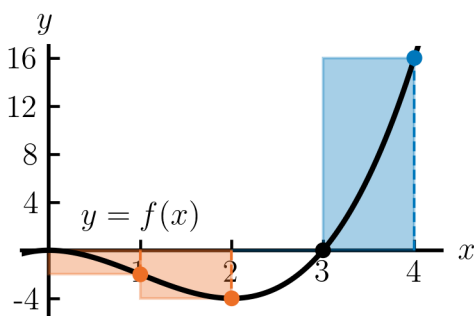


Definition Net Area: The net area bounded by the graph of f and the x -axis on $[a, b]$ is the area above the x -axis minus the area below the x -axis.

Example 1: Let $f(x) = x^3 - 3x^2$. Approximate the net area bounded by the graph of f and the x -axis on the interval $[0, 4]$ using 4 right-endpoint rectangles.

Solution: On $[0, 4]$ with $n = 4$ rectangles, the width is $\Delta x = \frac{b-a}{n} = \frac{4-0}{4} = 1$.

The right endpoints are $x_k^* = a + k\Delta x = 0 + k \cdot 1 = k$ for $k = 1, 2, 3, 4$.



$$\begin{aligned} \sum_{k=1}^4 \underbrace{f(k)}_{\text{height}} \cdot \underbrace{1}_{\text{width}} &= f(1) + f(2) + f(3) + f(4) \\ &= (1^3 - 3 \cdot 1^2) + (2^3 - 3 \cdot 2^2) + (3^3 - 3 \cdot 3^2) + (4^3 - 3 \cdot 4^2) \\ &= -2 - 4 + 0 + 16 = \boxed{10} \end{aligned}$$

Rectangles below the x -axis count as negative area. The net area is approximately $\boxed{10}$.

Using $n = 16$ right-endpoint rectangles gives a closer approximation of the net area.

The Riemann sum $\sum_{k=1}^n f(x_k^*) \Delta x$ approximates the net area bounded by the graph of f and the x -axis on $[a, b]$. Taking the limit as $n \rightarrow \infty$, the Riemann sum approaches the exact net area.

Definition Definite Integral: The definite integral of f on $[a, b]$ is defined by

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{k=1}^n f(x_k^*) \Delta x,$$

if the limit exists and is independent of the choice of x_k^* . If the definite integral exists, we say that f is **integrable** on $[a, b]$.

- a and b are the **limits of integration**
- $f(x)$ is the **integrand**
- x is the **variable of integration**

Theorem: If f is continuous on $[a, b]$, or is bounded with finitely many discontinuities on $[a, b]$, then f is integrable on $[a, b]$.

Example 2: The expression $\lim_{n \rightarrow \infty} \sum_{k=1}^n \left((x_k^*)^2 + 5x_k^* - 1 \right) \cdot \frac{3}{n}$ is a limit of a Riemann sum for a function f on $[0, 3]$. Find the function f and write the expression as a definite integral.

Solution: Compare the expression to the definition $\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{k=1}^n f(x_k^*) \Delta x$.

The width is $\Delta x = \frac{3}{n} = \frac{3-0}{n}$, which matches the interval $[0, 3]$.

The height is $f(x_k^*) = (x_k^*)^2 + 5x_k^* - 1$, so $f(x) = x^2 + 5x - 1$.

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n \left((x_k^*)^2 + 5x_k^* - 1 \right) \cdot \frac{3}{n} = \int_0^3 (x^2 + 5x - 1) dx$$

In the last example, we wrote a limit of a Riemann sum as a definite integral. Next, try this interactive problem on the website to go in the other direction: writing a definite integral as a limit of a Riemann sum.

Interactive Problem 1: Write the definite integral $\int_1^4 \sqrt{x} \, dx$ as a limit of a right-endpoint Riemann sum.

(a) The definition of the definite integral is $\int_a^b f(x) \, dx = \lim_{n \rightarrow \infty} \sum_{k=1}^n f(x_k^*) \Delta x$

(b) The width of each subinterval is $\Delta x = \frac{b-a}{n} = \frac{4-1}{n} = \frac{3}{n}$

(c) The formula for right-endpoints is $x_k^* = a + k\Delta x$

(d) The right-endpoints evaluate to $x_k^* = 1 + \frac{3k}{n}$

(e) Putting this all together,

$$\int_1^4 \sqrt{x} \, dx = \lim_{n \rightarrow \infty} \sum_{k=1}^n f(x_k^*) \Delta x = \lim_{n \rightarrow \infty} \sum_{k=1}^n \sqrt{1 + \frac{3k}{n}} \cdot \frac{3}{n}$$